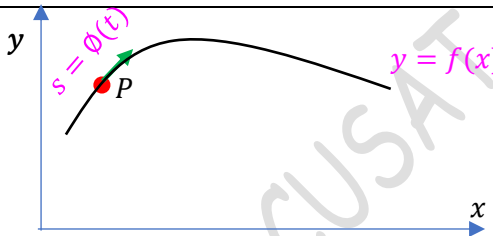
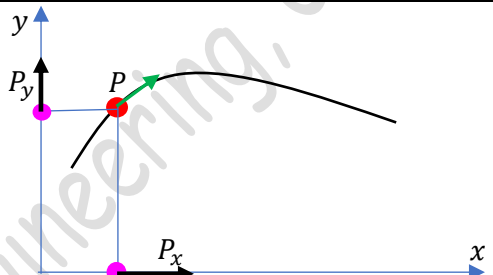


6. CURVILINEAR TRANSLATION

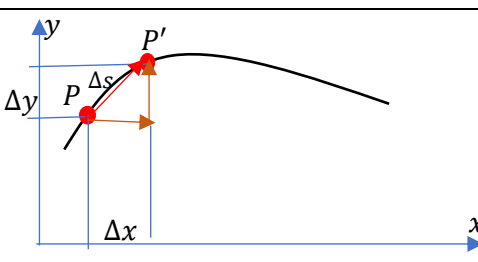
6.1 Kinematics of Curvilinear Translation

When a body moves along a curved path in such a way that each and every point in that body have the same displacement, velocity, acceleration etc., it is said to undergo a curvilinear translation.

Displacement

Consider a particle located at P on a curve defined by $y = f(x)$. We know that the position of the particle is changing continuously with respect to time. If we can represent the position s as function of t ($s = \phi(t)$), the equations together define the motion of the particle along the curve.	
We can also define the motion of a particle along a curve in another way. As the particle moves along the curve, both its x and y co-ordinates change with respect to time. If we express $x = f_1(t)$ and $y = f_2(t)$, then also the motion is completely defined. Let P_x and P_y denote the projections of the position P of the body moving along the curve on to the co-ordinate axes. As P moves along the curve both P_x and P_y change as functions of time. Hence, a curvilinear translation can be represented by two rectilinear translations of the projections of the body in the x-y axes.	

Velocity

Let the particle travels from P to P' in a time t to $t + \Delta t$, covering a distance Δs along the curve. The instantaneous velocity, $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$ In the limit, the direction of the instantaneous velocity will coincide with the tangent to the path at P corresponding to the instant t.	
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Hence, its direction continuously changes with time. To overcome this difficulty, consider the projection of Δs in the x and y directions.

$$\overrightarrow{\Delta s} = \overrightarrow{\Delta x} + \overrightarrow{\Delta y}$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\overrightarrow{\Delta s}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\overrightarrow{\Delta x} + \overrightarrow{\Delta y}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\overrightarrow{\Delta x}}{\Delta t} + \lim_{\Delta t \rightarrow 0} \frac{\overrightarrow{\Delta y}}{\Delta t}$$

$$\vec{v} = \vec{\dot{x}} + \vec{\dot{y}}$$

$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$

Acceleration

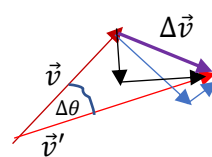
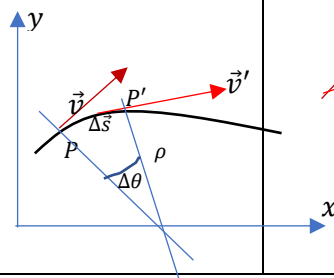
Let the velocity of the particle at P be $\vec{v}$ at any instant t . Velocity at P' in the next instant of time $t + \Delta t$ be $\vec{v}'$

$$\vec{v}' = \vec{v} + \Delta\vec{v}$$

The instantaneous acceleration

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t}$$

The direction of acceleration is in the direction of $\Delta\vec{v}$.



The direction of acceleration is continuously changing with time. We can resolve $\Delta\vec{v}$ in to two mutually perpendicular directions $\Delta\vec{v}_x$ and $\Delta\vec{v}_y$,

$$\begin{aligned}\Delta\vec{v} &= \Delta\vec{v}_x + \Delta\vec{v}_y \\ \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} &= \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_x + \Delta\vec{v}_y}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_x}{\Delta t} + \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_y}{\Delta t} \\ \vec{a} &= \vec{\ddot{x}} + \vec{\ddot{y}} \\ a &= \sqrt{\ddot{x}^2 + \ddot{y}^2}\end{aligned}$$

Since the velocity is tangential to the curve, the normal to the velocities will intersect at the centre of curvature subtending a small angle $\Delta\theta$. Let ρ be the radius of curvature.

$$\Delta s = \rho \Delta\theta$$

We can also resolve $\Delta\vec{v}$ into two mutually perpendicular components, along the normal and tangential to the curve at P.

$$\Delta\vec{v} = \Delta\vec{v}_n + \Delta\vec{v}_t$$

$$\Delta v_n = v \cdot \Delta\theta = v \cdot \frac{\Delta s}{\rho}$$

$$\Delta v_t \cong v' - v = \Delta v$$

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_n + \Delta\vec{v}_t}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_n}{\Delta t} + \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_t}{\Delta t} = \vec{a}_n + \vec{a}_t$$

$$a = \sqrt{a_n^2 + a_t^2}$$

$$\begin{aligned}\vec{a}_n &= \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_n}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{v \cdot \frac{\Delta s}{\rho}}{\Delta t} = \frac{v}{\rho} \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{v}{\rho} \cdot \frac{ds}{dt} = \frac{v^2}{\rho} \\ \vec{a}_n &= \frac{v^2}{\rho}\end{aligned}$$

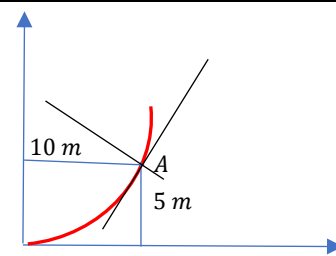
$$\begin{aligned}\vec{a}_t &= \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}_t}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \\ \vec{a}_t &= \frac{dv}{dt}\end{aligned}$$

Hence it can be seen that the tangential acceleration $\frac{dv}{dt}$ depends on the rate of change speed of the particle, while the normal acceleration depends on the square of the speed and the curvature of the path and is always directed towards the centre of curvature. When the particle moves along a straight path, the radius of curvature is ∞ and hence normal acceleration vanishes.

The tangential acceleration accounts for the change in the magnitude of the velocity whereas the normal acceleration accounts for the change in the direction of velocity.

Example 1

A particle travels with a constant speed of 6 m/s along the parabolic path $y = \frac{x^2}{20}$ as shown. Determine its acceleration at point A.



Since the particle is moving with constant speed, its tangential acceleration, $a_t = 0$

The normal acceleration, $a_n = \frac{v^2}{\rho}$

From calculus, we know

$$\text{curvature, } \frac{1}{\rho} = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{x}{10} \\ \frac{d^2y}{dx^2} &= \frac{1}{10} \end{aligned}$$

At A(10,5),

$$\begin{aligned} \frac{1}{\rho} &= \frac{1}{28.28} \text{ m}^{-1} \\ a_n &= \frac{6^2}{28.28} = 1.273 \text{ m/s}^2 \end{aligned}$$

The resultant acceleration, $a = \sqrt{a_n^2 + a_t^2} = 1.273 \text{ m/s}^2$ in the normal direction towards the centre of curvature of the path.

Example 2

A race car travels around a horizontal circular track that has a radius $\rho = 90 \text{ m}$. If the car increases its speed at a constant rate of 2.1 m/s^2 starting from rest, determine the time needed for it to reach a resultant acceleration of 2.4 m/s^2 . What is its speed at this instant?

$a_t = 2.1 \text{ m/s}^2$
Initial velocity, $u = 0$
Let the car attains the acceleration 2.4 m/s^2 in time t
Velocity of the body at this instant, $v = u + a_t t$
 $v = 2.1 t \text{ m/s}$

$$\begin{aligned} \text{Resultant acceleration, } a &= \sqrt{a_n^2 + a_t^2} \\ 2.4 &= \sqrt{a_n^2 + 2.1^2} \\ a_n &= 1.16 \text{ m/s}^2 \\ \frac{v^2}{\rho} &= 1.16 \lll v = 10.2 \text{ m/s} \\ \therefore t &= \frac{10.2}{2.1} = 4.87 \text{ s} \end{aligned}$$

#Exercise 1

Starting from rest, a particle moves along a circular path of radius r so that the distance travelled is given by the expression $s = ct^2$ where c is a constant. Find the normal and tangential components of acceleration?

#Exercise 2

A particle travels with a constant speed v along a parabolic path defined by the equation $y = kx^2$ where k is a constant. Find the maximum acceleration of the particle.

#Exercise 3

The curvilinear motion of a particle is defined by the displacement-time equation $x = v_0 t$ and $y = h \cos \frac{2\pi v_0 t}{l}$. What is the equation of the path $y = f(x)$. Find the maximum normal acceleration of the particle, if $v_0 = 10 \frac{\text{m}}{\text{s}}$, $h = 1 \text{ m}$, $l = 10 \text{ m}$

Prof. Biju N., School of Engineering, CUSAT

Prof. Biju N., School of Engineering, CUSAT